

Ordinary Differential Equations

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1 1st Order Linear ODE

$$(E) \dots a(x)y' + b(x)y = f(x)$$

$$y = y_h + y_p$$

$$y_h = Ce^{-\int \frac{b(x)}{a(x)} dx}$$

$$y_p = C(x)e^{-\int \frac{b(x)}{a(x)} dx}$$

$$C(x) = \int \frac{f(x)}{a(x)} e^{\int \frac{b(x)}{a(x)} dx} dx$$

2 1st Order Nonlinear ODE

2.1 Bernoulli

$$(E) \iff a(x)y' + b(x)y = y^n f(x), n > 1$$

$$(E) \iff a(x)y'y^{-n} + b(x)y^{-n} = f(x)$$

$$z = y^{1-n} \implies z' = (1-n)y^{-n}y'$$

$$(E) \iff a(x)z + (1-n)b(x)z = (1-n)f(x)$$

2.2 Riccati

$$(E) \iff a(x)y' + b(x)y = y^2 f(x) + g(x)$$

assume y_p a particular solution of (E)

$$\implies y = y_p + \frac{1}{z} \text{ with } z \text{ the solution of the homogenous equation } (E_h) \text{ such that } z = \frac{1}{y}$$

$$(E_h) \iff a(x)z' - b(x)z = -f(x)$$

3 2nd Order Linear ODE

$$(E) \iff ay'' + by' + cy = f(x)$$

First, we find its homogenous equation: $(E_h) \iff ay'' + by' + cy = 0$

Then, we find the characteristic equation: $(E_c) \iff ar^2 + br + c = 0$

We, then have to compute $\Delta = b^2 - 4ac$

- If $\Delta > 0$, have two solutions: $r_1, r_2 \implies y_h = Ae^{r_1x} + Be^{r_2x} \left| \begin{array}{l} y_1 = e^{r_1x} \\ y_2 = e^{r_2x} \end{array} \right.$
- If $\Delta = 0$, have double solution: $r_0 \implies y_h = (A + Bx)e^{r_0x} \left| \begin{array}{l} y_1 = e^{r_1x} \\ y_2 = xe^{r_2x} \end{array} \right.$
- If $\Delta < 0$, have two complex solutions: $r = \alpha \pm i\beta \implies y_h = e^{\alpha x}(A \cos \beta x + B \sin \beta x) \left| \begin{array}{l} y_1 = e^{\alpha x} \cos \beta x \\ y_2 = e^{\alpha x} \sin \beta x \end{array} \right.$

Now, we need to find the particular solution using Lagrange (variation of constants):

$$y_p = A(x)y_1 + B(x)y_2$$

To find $A(x)$ and $B(x)$ we solve the following linear system using carmer:

$$\begin{cases} A'(x)y_1 + B'(x)y_2 = 0 \\ A'(x)y_1' + B'(x)y_2' = \frac{f(x)}{a} \end{cases} \iff \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} A'(x) \\ B'(x) \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{f(x)}{a} \end{pmatrix}$$

$$\implies \begin{cases} A'(x) = \frac{\begin{vmatrix} 0 & y_2 \\ \frac{f(x)}{a} & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} \\ B'(x) = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & \frac{f(x)}{a} \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} \end{cases}$$

Then, we integrate them and replace them in y_p

Then, the solution of (E) is $y = y_h + y_p$

Look up: Algebra 02: Carmer's

3.1 Cases of $f(x)$

If $f(x) = P_n e^{\alpha x} | P_n \in \mathbb{R}_n[X]$, Then the solution is

- if α is a simple solution of (E_c) then $y = xQ_n e^{\alpha x}$

- if α is a double solution of (E_c) then $y = x^2 Q_n e^{\alpha x}$

- if α is not a solution of (E_c) then $y = Q_n e^{\alpha x}$

If $f(x) = e^{\alpha x}(P_1 \cos \beta x + P_2 \sin \beta x)$, $P_1, P_2 \in \mathbb{R}_n[X]$, Then the solution is

- if $\alpha + i\beta$ is a solution of $(E_c) \implies y = x e^{\alpha x}(Q_1 \cos \beta x + Q_2 \sin \beta x)$

- if $\alpha + i\beta$ is not a solution of $(E_c) \implies y = e^{\alpha x}(Q_1 \cos \beta x + Q_2 \sin \beta x)$

3.2 Remark

If

$$(E) \iff ay'' + by' + cy = \sum_{k=1}^n f_k = f_1(x) + f_2(x) + \dots + f_n(x)$$

or

$$(E) \iff a(x)y' + b(x)y = \sum_{k=1}^n f_k$$

Then we first find the homogenous solution as usual then we find the particular solution of the equations y_{p_k}

$$ay'' + by' + cy = f_k(x)$$

Or

$$a(x)y' + b(x)y = f_k(x)$$

using the previous methods. Then, the solution is: $y = y_h + \sum_{k=1}^n y_{p_k}$

3.3 Remark: Cauchy

Cauchy aims to find one solution to the differential equation i.e replacing the constants of y_h with actual numbers

If the ODE is of 1st order then the condition given is: $y(x_0) = y_0$

If the ODE is of 2nd order then the condition given is: $\begin{cases} y(x_0) = y_0 \\ y'(x_0) = y'_0 \end{cases}$